A local power-series obstruction to a C*-algebraic Corona conjecture

Research proof and Lean formalization

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Abstract

Under the local left-coefficient power-series definition used in the source, a uniformly invertible affine function on the norm unit ball of $M_2(\mathbb{C})$ need not have an analytic reciprocal in that class. A square-zero matrix reduces the obstruction to two finite polynomial identities. This refutes the source's C*-algebraic Corona assertion already for one datum. Ordinary complex and operator-valued holomorphic Corona results are unaffected.

Krishna's Definition 2.6 in [1] calls a function analytic when, near every center a, it has a norm-convergent expansion

$$f(z) = \sum_{n\geq 0} b_n (z-a)^n$$

with coefficients on the left. Conjecture 3.2 asserts that bounded functions $f_1, \dots, f_k$ in this class on the norm-open unit ball of a unital C*-algebra, satisfying

$$\sum_j f_j(z) f_j(z)^* \geq \delta I, \qquad \sum_j f_j(z)^* f_j(z) \geq \delta I$$

uniformly for some $\delta > 0$, admit bounded functions g_j, h_j of the same class with $\sum_j f_j g_j = I = \sum_j h_j f_j$. No commutativity assumption is imposed.

Theorem 1. *Let $\mathcal{A} = M_2(\mathbb{C})$ with its operator norm, let*

$$c = \begin{pmatrix} 1/4 & 0 \\ 0 & 0 \end{pmatrix}, \qquad f(z) = I + cz.$$

On $\{z : \|z\| < 1\}$, the function f is bounded and analytic in the source's sense, and satisfies both lower bounds with $\delta = 1/2$. There is no function g having a source power-series expansion even at 0 and satisfying $f(z)g(z) = I$ throughout that ball.

Proof. At any center a, the finite expansion $f(z) = f(a) + c(z-a)$ proves the required analyticity. Since $\|c\| = 1/4$, we have $\|f(z)\| \leq 5/4$. Set $T = cz$. The self-adjoint operator $T + T^*$ is bounded below by $-2\|T\| I$, and TT^*, T^*T are positive. Thus

$$(I+T)(I+T)^* \geq (1 - 2\|T\|)I \geq \tfrac{1}{2} I, \qquad (I+T)^*(I+T) \geq \tfrac{1}{2} I.$$

Here $\|T\| < 1/4$ for every input in the ball. In fact the ordinary reciprocal exists and is uniformly bounded by $4/3$, by the Neumann series.

Suppose $f(z)g(z) = I$ and $g(z) = \sum_{n\geq 0} b_n z^n$ on some ball about 0. Substitution of 0 gives $b_0 = g(0) = I$. Put

$$N = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Then

$$N^2 = 0, \qquad cN = \begin{pmatrix} 1/4 & 1/4 \\ 0 & 0 \end{pmatrix}, \qquad (cN)^2 = \begin{pmatrix} 1/16 & 1/16 \\ 0 & 0 \end{pmatrix} \neq 0. \tag{1}$$

Choose a sufficiently small real $t > 0$ so that both tN and $-tN$ lie in the expansion ball and the unit ball. Since all powers of either input of order at least two vanish, norm convergence gives exactly

$$g(tN) = I + tb_1 N, \qquad g(-tN) = I - tb_1 N.$$

Expanding the two right-inverse identities therefore yields

$$t(cN + b_1 N) + t^2 cN b_1 N = 0, \qquad -t(cN + b_1 N) + t^2 cN b_1 N = 0.$$

Subtracting gives $b_1 N = -cN$; adding gives $cN b_1 N = 0$. Consequently $(cN)^2 = 0$, contradicting (1). No differentiation or interchange of an infinite series is needed. □

Scope. Theorem 1 contradicts the conclusion of Conjecture 3.2 at $k = 1$: even its right solution cannot exist in the required class. The obstruction concerns the prescribed local powers in a noncommutative algebra. The ordinary Fréchet-holomorphic reciprocal exists; there is no contradiction with scalar Carleson theory or with a different notion of noncommutative analyticity. Rosenfeld's matrix-algebra rigidity theorem already restricts the relevant differentiable class to affine maps [3, Theorem 2]. Restrictions on products are also discussed in [2, Theorem 5.10].

Formal scope and status. The closed Lean endpoint `CStarCorona.source_counterexample` refutes the literal matrix-instance assertion, using the actual operator norm, positive-semidefinite order and local norm-convergent left-power series. The parent strict replay passes 25 declarations in four modules under Lean 4.34.1 and mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`,

with only `propext`, `Classical.choice` and `Quot.sound`. The ordinary Neumann reciprocal observation is classical background.

Discovery credit is zero: independent preliminary review confirms the mathematics but rejects substantive eligibility as an immediate consequence of classical rigidity. The later Lean proof and this note were not independently reviewed. No new analytic machinery or worldwide priority is claimed. The actual source was read in the author's Zenodo deposit; direct Preprints PDF access remains unavailable.

References

[1] K. Mahesh Krishna, *C*-algebraic Bieberbach, Robertson, Lebedev–Milin, Zalcman, Krzyz and Corona Conjectures*, Preprints (2022), v1, Definition 2.6 and Conjecture 3.2, pp. 15–16. doi:10.20944/preprints202208.0140.v1. Author deposit: Zenodo 6917925.

[2] J. S. Cook, *Introduction to $\mathcal{A}$-Calculus*, arXiv:1708.04135 (2017).

[3] B. Rosenfeld, *Differentiable functions in associative and alternative algebras and smooth surfaces in projective spaces over these algebras*, Publ. Inst. Math. (Beograd), N.S. **68**(82) (2000), 67–71.